



MUHAMMAD ABDULLAH

Mechatronics Engineer

ABOUT ME

Motivated and detail-oriented student passionate about robotics and technology. Proficient in creating practical projects, such as line following robots, and experienced with microcontrollers, sensors, and programming. Keen to explore new technologies and apply problem solving skills to real world challenges.

EDUCATION

Matric

Dar E Arqam School
Marks: 1051/1100
Grade: A+

Intermediate

Pak Garrison College
Marks: 1055/1200
Grade: A+

Mechatronics Engineering

In Progress - 2029

SKILLS

Arduino Programming	
Basic Python Programming	
AutoCAD (2D/3D Design)	
Sensor Integration	
Motor Control Systems	
Electrical Analysis	
C++	

LANGUAGE

- English
- Urdu



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[muhammad-abdullah-portfolio](#)



Islamabad, Pakistan

PROJECTS



Line Follower Robot

2025

Robot Automation I Technology

- The line follower robot is an autonomous system designed to detect and follow a specific path using infrared sensors. It continuously reads surface contrasts and adjusts motor speeds to maintain direction. This project involves microcontrollers, motor drivers, and real time programming. It helps in understanding robotics concepts, sensor calibration, and embedded system design effectively.



Robot Gripper Mechanism

2025

AutoCAD I Mechanical Design

- The robot gripper mechanism is a mechanical design created using AutoCAD to simulate object handling in robotic systems. It focuses on precise movement and firm gripping using joints and linkages. The design helps in understanding mechanical structures, force distribution, and CAD modeling. This project improves skills in 2D/3D design and practical engineering concepts.



Grocery Price Prediction & Weekly Budget Planning

2025

Application I Budget Planner

- The Grocery Price Prediction and Weekly Budget Planning system is designed to estimate future grocery prices using available online data and assist in managing weekly expenses. It analyzes price trends and helps users plan their budget efficiently. The project focuses on data analysis, prediction techniques, and financial planning.



Smart Home Automation System

2026

IoT-Based Control System

- The Smart Home Automation System is designed to enable remote and automated control of household devices using IoT technology. It allows users to manage appliances such as lights, fans through a voice or web interface. The system enhances convenience, energy efficiency, and security by integrating sensors, microcontrollers, and real time monitoring. It also includes an authentication feature that requires a password for access, ensuring only authorized users can control the system. The project focuses on IoT development, embedded systems, and intelligent home control solutions.

○ Rectifier with RLC Filter Project

Power Electronics Application

- The AC to DC Rectifier with RLC Filter project is designed to convert alternating current (AC) into direct current (DC) using a rectifier circuit and smooth the output using an RLC filter. The rectifier changes AC input into pulsating DC, while the RLC filter reduces ripples and provides a stable DC output. The system improves power quality and efficiency for electronic devices that require steady DC supply. The project focuses on power electronics, circuit design, filtering techniques, and practical applications in regulated power supplies.
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